

ENTERPRISE INFRASTRUCTURE, SECURITY & OPERATIONAL SUPPORT STRATEGY

IT Support Professional
Certificate Capstone Project

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Tech Career Accelerator
Google IT Support Capstone Project

Table of Contents

Table of Contents	1
Executive Summary	5
Business Requirements Analysis	7
Company Overview.....	7
Business Objectives	7
Developer Productivity	7
Operational Continuity	7
Intellectual Property Protection	8
Cost Efficiency	8
Scalability and Growth.....	8
Departmental Requirements	8
Executive Management.....	8
Finance Department	8
Human Resources Department	8
Sales and Marketing Department	9
Software Development Department	9
IT Support and Administration.....	9
Technical Requirements	9
Compute Performance.....	9
Centralized Data Management.....	9
High-Speed Networking	9
Security Controls.....	10
Business Continuity.....	10
Scalability	10
Reliability and Availability.....	10
Infrastructure Design.....	11
Hardware Inventory	11
Software Inventory	12
Finance Department Software Deployment.....	13

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Backup Strategy	29
Backup Infrastructure	29
Network Attached Storage (NAS)	29
Cloud Backup Storage	30
Backup Schedule	30
Daily Incremental Backups	30
Weekly Full Backups	31
Monthly Archive Backups	31
Offsite Cloud Storage	31
Cloud Security Controls	32
Recovery Objectives	32
Recovery Time Objective (RTO)	32
Recovery Point Objective (RPO)	33
Disaster Recovery Procedures	33
Cybersecurity Framework	35
Cybersecurity Policy	35
Password Complexity Requirements	35
Device Encryption	36
Patch Management	36
Deployment timelines:	37
Acceptable Use Policy	37
Risk Assessment Matrix	37
Incident Response Plan	38
Detection	38
Reporting	38
Containment	39
Eradication	39
Recovery	40
Lessons Learned	40
Security Awareness Training Program	41
Training Module 1: Phishing Awareness	41
Training Module 2: Password Security and Identity Protection	41

Training Module 3: Safe Internet Usage	42
Training Module 4: Device and Physical Security	42
Training Module 5: Incident Reporting	42
Training Program Summary	43
Technical Support Operations	44
Workstation Power Failure	44
Network Connectivity Failure	45
Shared Printer Failure	45
Project Outcomes & Recommendations	47
Project Outcomes	47
Conclusion	50
References	51

Executive Summary

This report presents a comprehensive IT infrastructure solution developed for Ubuntu Innovations (Pty) Ltd, a growing software development and technology consulting company with 25 employees across Executive Management, Finance, Human Resources, Sales and Marketing, and Software Development. The proposed solution addresses the organisation's operational, technical, and security requirements while supporting future business growth, improved productivity, and enhanced cybersecurity.

The project begins with a detailed business requirements analysis to identify the company's operational objectives, technical needs, security considerations, and future scalability requirements. Based on these findings, appropriate hardware and software solutions are recommended, including enterprise-grade workstations, networking equipment, storage infrastructure, and department-specific software designed to improve efficiency and reliability.

A secure network architecture and segmented IP addressing plan have been developed to support effective communication, simplify network management, and strengthen security. Role-based access control is implemented through a User and Permissions Matrix, structured shared folder design, and operating system administration procedures to ensure employees can access only the resources required for their roles.

Business continuity is supported through a comprehensive Backup and Disaster Recovery Plan incorporating daily incremental backups, weekly full backups, monthly archive backups, secure offsite cloud storage, and clearly defined Recovery Time and Recovery Point Objectives. A Corporate Cybersecurity Framework further strengthens the organisation by establishing security policies covering password management, multi-factor authentication, endpoint protection, device encryption, patch management, acceptable use, risk management, incident response, and employee security awareness training.

To support day-to-day IT operations, the report also includes practical troubleshooting guides for common technical issues encountered within business environments. Finally, career readiness materials—including a professional résumé, cover letter, LinkedIn profile summary, and interview preparation resources—demonstrate the practical skills developed throughout the project.

Overall, this report provides Ubuntu Innovations (Pty) Ltd with a secure, scalable, and professionally documented IT infrastructure solution that aligns with industry best practices and demonstrates the technical,

analytical, and documentation skills expected of an entry-level IT Support Specialist.

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Business Requirements Analysis

Company Overview

Ubuntu Innovations is a fast-growing, dynamic software development and technology consulting firm. The company specializes in building modern web applications, cloud-native solutions, and custom software integrations for local and global enterprises. Grounded in the philosophy of collaborative innovation, the organization relies heavily on robust, high-availability developer environments, seamless cross-team communication, and secure handling of intellectual property, including source code, proprietary client data, and development environments.

The company currently employs 25 staff members across five key departments: Executive Management, Finance, Human Resources, Sales and Marketing, and Software Development. As the organisation continues to grow, it requires a secure and scalable technology infrastructure capable of supporting daily operations while maintaining high levels of performance, security, and reliability.

To achieve these objectives, Ubuntu Innovations requires modern hardware, secure networking infrastructure, centralized data storage, structured access controls, backup solutions, and cybersecurity measures that support both current business requirements and future expansion.

Business Objectives

The primary objective of this project is to design a reliable and secure IT infrastructure that supports the operational needs of Ubuntu Innovations while providing a foundation for future growth.

The business objectives of the organisation include:

Developer Productivity

Provide employees, particularly software developers, with modern computing resources capable of supporting software development tools, virtual environments, integrated development environments (IDEs), and containerized applications. This will reduce delays caused by hardware limitations and improve overall productivity.

Operational Continuity

Implement reliable infrastructure solutions that minimise downtime and ensure employees can continue working efficiently. Systems must be designed to support business continuity through redundancy, backup procedures, and effective disaster recovery strategies.

Intellectual Property Protection

Protect sensitive company information, including source code repositories, customer data, employee records, financial information, and business documentation. Security controls must prevent unauthorized access and reduce the risk of data breaches.

Cost Efficiency

Select hardware, software, and cloud services that provide long-term value while balancing performance, reliability, and cost. The infrastructure should minimise unnecessary expenditure while meeting operational requirements.

Scalability and Growth

Design solutions that can support future business expansion without requiring significant infrastructure replacement. Systems should accommodate additional users, devices, applications, and storage requirements as the organisation grows.

Departmental Requirements

Each department within Ubuntu Innovations has specific operational and technology requirements that must be supported by the proposed infrastructure.

Executive Management

Management requires secure access to company reports, strategic planning documentation, financial summaries, contracts, and executive communications. Management staff require access to business intelligence information while maintaining appropriate security controls.

Finance Department

The Finance Department requires secure storage and access to payroll information, accounting records, budgeting documentation, tax records, financial statements, and accounting software. Due to the sensitive nature of financial information, strict access controls and backup procedures are required.

Human Resources Department

Human Resources requires secure management of employee records, recruitment documentation, leave records, performance reviews, contracts, and compliance documentation. Access to HR information must be restricted to authorized personnel only.

Sales and Marketing Department

The Sales and Marketing team requires access to customer relationship management systems, marketing materials, sales reports, campaign data, presentations, and customer communications. Reliable internet access and collaboration tools are essential to support daily operations.

Software Development Department

The Software Development Department requires powerful workstations, source code repositories, development environments, testing platforms, databases, version control systems, and collaboration tools. This department generates significant amounts of data and relies heavily on secure, high-performance infrastructure.

IT Support and Administration

IT personnel require administrative access to company systems, network infrastructure, backup solutions, security tools, and monitoring platforms. They are responsible for maintaining infrastructure availability, managing user accounts, supporting employees, and ensuring compliance with security policies.

Technical Requirements

To support the operational needs of Ubuntu Innovations, the proposed infrastructure must meet several technical requirements relating to performance, security, availability, and scalability.

Compute Performance

Employees require modern computing devices capable of supporting multitasking, productivity software, communication platforms, web applications, and department-specific tools. Software developers require additional processing power and memory to support development environments, virtual machines, and containerized applications.

Centralized Data Management

The organisation requires centralized storage solutions that provide secure access to company files, shared resources, backup repositories, and departmental data. Centralized storage improves collaboration, simplifies administration, and enhances data protection.

High-Speed Networking

The network infrastructure must provide reliable connectivity for both wired and wireless devices. Employees require consistent access to business

applications, cloud services, file storage systems, printers, and communication platforms. The network must support future growth while maintaining high performance levels.

Security Controls

The infrastructure must incorporate strong security controls including firewalls, endpoint protection, encryption, multi-factor authentication, access control policies, and secure backup solutions. These controls are necessary to protect company information from internal and external threats.

Business Continuity

Systems must support backup and disaster recovery procedures that minimise downtime and reduce the impact of data loss incidents. Critical business information must be recoverable within defined recovery objectives.

Scalability

The infrastructure must support organisational growth by allowing additional users, devices, storage capacity, software services, and network resources to be added without major redesign or replacement of existing systems.

Reliability and Availability

Business systems must remain available and operational during normal business activities. Hardware, networking equipment, and software solutions should be selected based on their reliability, supportability, and long-term sustainability.

These technical requirements form the foundation upon which the remainder of the proposed infrastructure solution has been designed.

Infrastructure Design

Hardware Inventory

Ubuntu Innovations (Pty) Ltd requires reliable, secure, and high-performance hardware capable of supporting daily business operations, software development activities, data storage requirements, and future organisational growth. The recommended hardware has been selected based on performance, reliability, scalability, security features, and overall cost effectiveness.

The recommended employee workstation is the Lenovo ThinkPad T14s Gen 5 equipped with an AMD Ryzen 7 PRO processor, 32GB of RAM, and a 1TB NVMe SSD. This configuration provides sufficient processing power to support multitasking, software development environments, productivity applications, virtual machines, and cloud-based services. The ThinkPad range is widely recognised for its reliability, durability, and long lifecycle, making it suitable for business use.

To support centralized file storage and backup operations, the organisation will deploy a Synology DiskStation DS1823xs+ Network Attached Storage (NAS) device. The NAS will be configured using RAID 6 technology to provide redundancy and protect against drive failures. This solution offers secure file storage, automated backups, centralized management, and high-speed access to company resources.

Network infrastructure will consist of a UniFi Dream Machine SE, UniFi Switch Pro 24 PoE, and UniFi U6 Pro Wireless Access Points. These devices provide enterprise-grade networking capabilities including firewall protection, centralized management, Power over Ethernet support, wireless connectivity, and advanced security controls.

Shared printing requirements will be supported through HP LaserJet Pro MFP M428fdn multifunction printers. These devices provide reliable printing, scanning, and document management capabilities while maintaining low operational costs and high availability.

The selected hardware infrastructure provides Ubuntu Innovations with a scalable and reliable platform capable of supporting current business requirements while allowing future expansion.

Table 1: Hardware Inventory summarises the recommended hardware required to support the operational, networking, storage, and security requirements of Ubuntu Innovations (Pty) Ltd.

Component	Recommended Device	Purpose
Employee Workstations	Lenovo ThinkPad T14s Gen 5	Daily business operations
NAS Storage	Synology DS1823xs+	Backup and file storage
Router/Firewall	UniFi Dream Machine SE	Security and routing
Managed Switch	UniFi Switch Pro 24 PoE	Network connectivity
Wireless Access Points	UniFi U6 Pro	Wireless networking
Printers	HP LaserJet Pro M428fdn	Printing and scanning

Table 1: Hardware Inventory

Software Inventory

The software inventory has been designed to support productivity, collaboration, security, administration, software development, and business operations across all departments within Ubuntu Innovations.

Standard software deployed to all users includes Windows 11 Pro, Microsoft 365, Microsoft Defender, Adobe Acrobat Reader, and web browsers such as Google Chrome or Microsoft Edge. These applications support communication, document management, internet access, and endpoint security.

Department-specific software is deployed according to operational requirements. Finance personnel utilize Sage Accounting and Payroll software for financial management and payroll processing. Human Resources uses BambooHR for employee administration and recruitment activities. Sales and Marketing personnel utilize HubSpot CRM and Canva Pro to manage customer relationships and create marketing content.

Software developers are provided with Visual Studio Code, GitHub Desktop, Python, MySQL Workbench, PostgreSQL, MongoDB Compass, GitHub Copilot, and ChatGPT to support software development, database administration, testing, debugging, and collaboration.

IT administrators utilize specialized tools including TeamViewer, Jira Service Management, Wireshark, and Veeam Backup & Recovery to support system administration, troubleshooting, backup management, and technical support operations.

The selected software solutions provide a secure, productive, and scalable environment capable of supporting all business functions while maintaining licensing compliance and operational efficiency.

Table 2: Company-wide Software Inventory outlines the deployment strategy, installation methods, licensing requirements, and update procedures for software used throughout the organisation.

Software	Users	Deployment Method	License Type	Update Method
Windows 11 Pro	25	Pre-Installed on company devices	Per-Device License	Automatic Windows Updates
Microsoft 365	25	Cloud-based deployment through company accounts	Business Subscription	Automatic Cloud Updates
Microsoft Defender	25	Enabled during workstation configuration	Included with Windows	Automatic Security Updates
Web browser: Google Chrome / Microsoft Edge	25	Installed during workstation setup	Free	Automatic Browser Updates
Adobe Acrobat Reader	25	Installed during workstation setup	Free	Automatic Updates
Microsoft Entra ID (Azure AD)	25	Configured through company domain accounts	Business Subscription	Managed Cloud Updates

Table 2: Company-wide Software Inventory

Table 3: Finance Department Software Inventory-Table 7: IT Support & Admin Department Software Inventory presents the specialised software required by each department to support their operational responsibilities.

Finance Department Software Deployment

Software	Users	Deployment Method	License Type	Update Method	Purpose
Sage Accounting Payroll	4	Cloud-based deployment through secure company accounts	Business Subscription	Automatic Cloud Updates	Financial management, payroll processing, and tax calculations

Table 3: Finance Department Software Inventory

Human Resources Department

Software	Users	Deployment Method	License Type	Update Method	Purpose
BambooHR	3	Cloud-based deployment through company accounts	Business Subscription	Automatic Cloud Updates	Employee records, leave management, and recruitment tracking

Table 4: HR Department Software Inventory

Sales and Marketing Software Deployment

Software	Users	Deployment Method	License Type	Update Method	Purpose
Canva Pro	7	Cloud-based deployment through team accounts	Team Subscription	Automatic Cloud Updates	Marketing material and presentation design
HubSpot CRM	7	Cloud-based deployment through company accounts	Subscription	Automatic Cloud Updates	Manages customer relationships, tracks sales activities, monitors leads, and supports marketing campaigns.

Table 5: Sales & Marketing Department Software Inventory

Software Development Department Software Deployment

Software	Users	Deployment Method	License Type	Update Method	Purpose
Visual Studio Code	8	Installed on development workstations	Free / Open Source	Automatic Updates	Source code editing, debugging, application development, and software testing.
GitHub Desktop	8	Installed on development workstations	Free	Automatic Updates	Version control management and collaboration on software projects.
Python	8	Installed on development workstations	Free / Open Source	Manual and Automatic Updates	Application development, automation scripting, testing, and data processing.
MySQL Workbench	8	Installed on development workstations	Free / Open Source	Automatic Updates	Database design, SQL development, and database administration.
PostgreSQL	8	Installed on development workstations	Free / Open Source	Scheduled Updates	Enterprise database management and application data storage.
MongoDB Compass	8	Installed on development workstations	Free	Automatic Updates	Management and administration of NoSQL databases.
GitHub Copilot	8	Configured through GitHub accounts	Subscription	Managed Cloud Updates	AI-assisted coding, code suggestions, and developer productivity enhancement.

ChatGPT	8	Cloud-based access through company accounts	Subscription / Free	Managed Cloud Updates	Assists with coding support, documentation, troubleshooting, research, and productivity.
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Table 6: Software Department Software Inventory

IT Support and Administration Tools

Software	Users	Deployment Method	License Type	Update Method	Purpose
TeamViewer	3	Installed on IT support devices	Subscription	Automatic Updates	Remote troubleshooting, user support, and system maintenance.
Jira Service Management	3	Cloud-based deployment through IT accounts	Subscription	Automatic Cloud Updates	Incident management, service requests, ticket tracking, and IT support operations.
Wireshark	3	Installed on IT administration devices	Free / Open Source	Manual Updates	Network analysis, packet inspection, and troubleshooting network issues.
Veeam Backup & Recovery	3	Installed on backup management systems	Subscription	Scheduled Updates	Backup management, disaster recovery, and business continuity support.

Table 7: IT Support & Admin Department Software Inventory

Software Installation Procedures

Ubuntu Innovations (Pty) Ltd will follow a structured deployment process to ensure software consistency, security, and compliance across all departments.

The deployment process includes:

- Company devices will be configured with Windows 11 Pro before being assigned to employees.
- Standard software applications will be installed during workstation setup.
- Department-specific software will be deployed according to employee roles and responsibilities.
- Employees will receive company-managed user accounts and credentials.

- Cloud-based applications will be configured using company email accounts.
- IT support personnel will verify software functionality before devices are issued.
- All software installations and licenses will be documented for auditing and maintenance purposes.

Software Update and Patch Management

To maintain security and system reliability, Ubuntu Innovations (Pty) Ltd will implement the following update procedures:

- Automatic Windows Updates will be enabled on all devices.
- Antivirus definitions and security signatures will update automatically.
- Cloud-based platforms such as Microsoft 365, HubSpot CRM, BambooHR, and Canva Pro will receive vendor-managed updates.
- Critical security updates will be deployed immediately by IT support staff.
- Unsupported or outdated software will be removed from company systems.
- Scheduled maintenance reviews will verify software compliance and update status.

Licensing Management

Ubuntu Innovations (Pty) Ltd will maintain valid licenses and subscriptions for all commercial software used within the organisation.

Commercial software requiring licensing includes:

- Windows 11 Pro
- Microsoft 365 Business
- Sage Accounting & Payroll
- BambooHR
- Canva Pro
- HubSpot CRM
- GitHub Copilot
- TeamViewer

- Jira Service Management
- Veeam Backup & Recovery

Open-source and free software used by the organisation includes:

- Visual Studio Code
- Python
- PostgreSQL
- MySQL Workbench
- Wireshark
- GitHub Desktop
- MongoDB Compass

Software Deployment Plan Conclusion

The Software Deployment Plan provides Ubuntu Innovations (Pty) Ltd with a structured approach to deploying, managing, updating, and securing software across all departments. Standardised software deployment improves productivity, strengthens cybersecurity, supports regulatory compliance, and ensures employees have access to the tools required to perform their roles effectively. By implementing controlled installation procedures, update management processes, and licensing controls, the organisation can maintain a secure, reliable, and efficient technology environment.

Network Architecture

Ubuntu Innovations (Pty) Ltd requires a secure, reliable, and scalable network infrastructure capable of supporting 25 employees across multiple departments. The network has been designed using a centralized hierarchical architecture that separates external internet connectivity from internal business resources while maintaining efficient communication between users, servers, printers, storage devices, and wireless devices.

The design follows a North-to-South traffic flow model, where all external traffic enters through a secure gateway before reaching internal network resources. This approach improves security, simplifies network management, and provides a clear structure for future growth.

The network consists of the following core components:

- Internet Service Provider (ISP) Fibre Connection
- Integrated Router and Firewall
- Core Managed Network Switch
- Employee Workstations
- Network Attached Storage (NAS)
- Shared Network Printers
- Wireless Access Points (WAPs)
- Staff Laptops and Mobile Devices

All wired devices are connected using Cat6 Ethernet cabling, while wireless devices connect through WPA3 Enterprise secured wireless access points.

Figure 1: Network Diagram illustrates the proposed network topology, showing how the organisation's infrastructure connects from the Internet Service Provider through the firewall and managed switch to all internal devices and services.

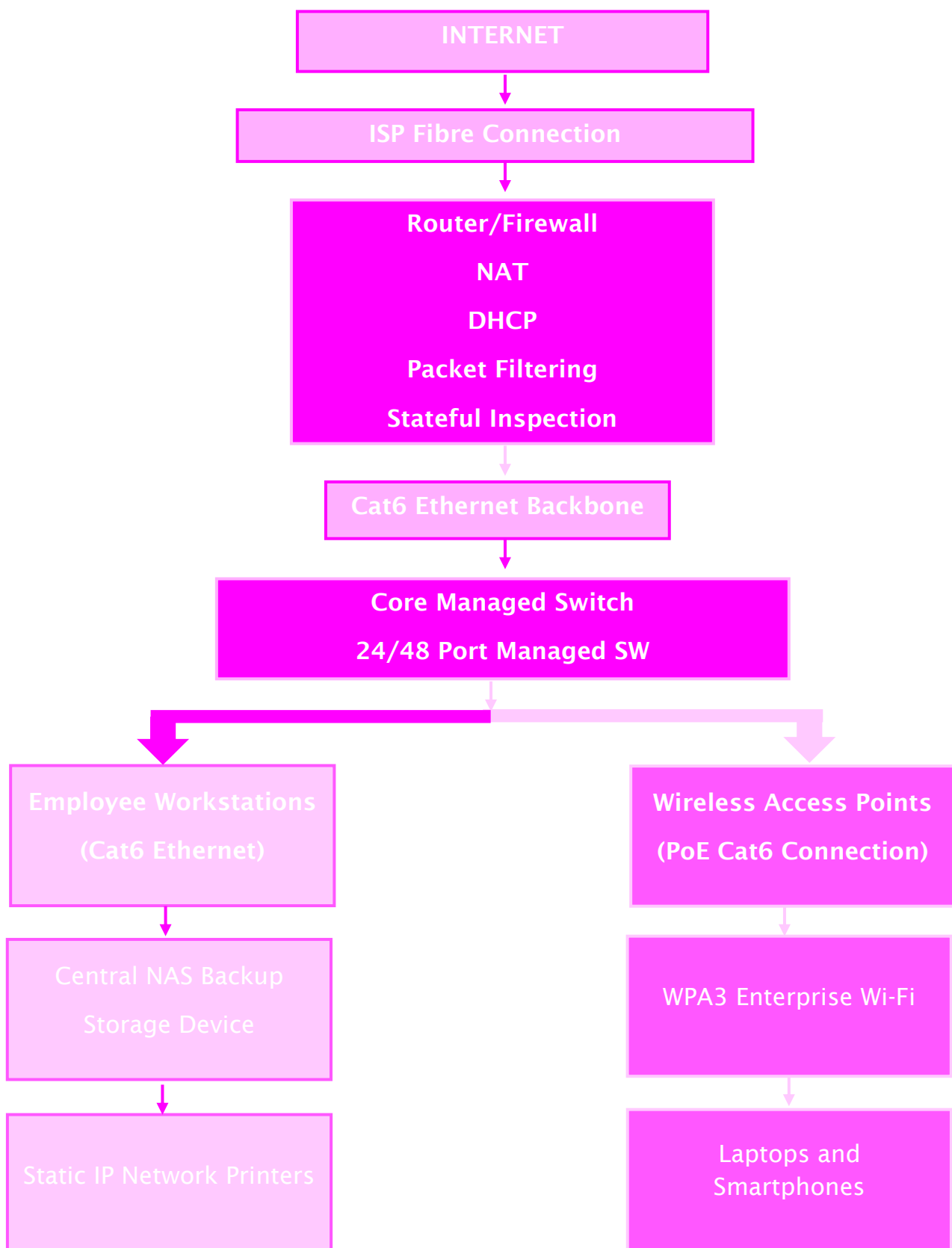


Figure 1: Network Diagram

Network Design Explanation

Ubuntu Innovations uses a centralized network design where all network traffic flows through a single security gateway before reaching internal resources. As illustrated in [Figure 1: Network Diagram](#), the network is designed using a hierarchical topology that improves performance, simplifies management, and strengthens security.

The Internet Service Provider (ISP) delivers internet connectivity through a high-speed fibre connection. This connection terminates at the Integrated Router and Firewall, which acts as the primary security gateway for the organisation.

The Router and Firewall performs several critical functions:

- Network Address Translation (NAT)
- DHCP Address Allocation
- Packet Filtering
- Stateful Inspection
- Internet Traffic Routing

A Cat6 Ethernet backbone connects the Router and Firewall to the Core Managed Switch. The Core Managed Switch functions as the central distribution point for all internal network traffic.

Connected to the switch are:

- Employee desktop workstations
- Central NAS backup storage
- Shared network printers
- Wireless access points

Wireless access points provide secure Wi-Fi connectivity to staff laptops and smartphones using WPA3 Enterprise security protocols.

This design provides reliable communication, centralized management, enhanced security, and scalability for future expansion.

IP Addressing Plan

Ubuntu Innovations requires a structured IP addressing scheme to improve network organisation, simplify troubleshooting, and support future growth.

The organisation will implement subnet separation based on device categories to improve network management and security.

Table 8: IP Addressing Table shows the proposed IP addressing scheme used to segment network traffic and improve security, performance, and troubleshooting.

Device Type	Subnet
Servers and NAS Storage	192.168.10.0/24
Employee Devices	192.168.20.0/24
Network Printers	192.168.30.0/24
Wireless Devices (Wi-Fi)	192.168.40.0/24

Table 8: IP Addressing Table

The proposed addressing structure separates critical infrastructure from user devices and printers. This approach simplifies device identification, improves troubleshooting efficiency, and provides a foundation for future network segmentation if the organisation expands.

System Administration & Access Management

Ubuntu Innovations requires a structured access management framework to ensure employees can securely access the resources required for their roles while protecting sensitive business information.

The organisation will implement role-based access control (RBAC) using department-specific security groups and shared folders. Access permissions will be assigned according to employee responsibilities, ensuring users receive only the minimum level of access necessary to perform their duties.

This approach supports:

- Improved data security
- Controlled access to sensitive information
- Easier administration of permissions
- Regulatory compliance
- Reduced risk of unauthorized access

The following sections outline the User and Permissions Matrix, Shared Folder Structure, Access Controls, Operating System Administration Guide, and Software Deployment Plan.

Table 9: User & Permissions presents the User and Permissions Matrix used to implement role-based access control across all departments.

User and Permissions Matrix

Department	Security Group	Folder Access
Management	mgmt_grp	Full Access
Finance	finance_grp	Finance Folder
Human Resources	hr_grp	HR Folder
Sales & Marketing	sales_grp	Sales Folder
Software Development	dev_grp	Software Software Development Folder

Table 9: User & Permissions

The User and Permissions Matrix establishes role-based access controls for Ubuntu Innovations (Pty) Ltd. Each department is assigned a dedicated security group that controls access to departmental resources and shared folders.

This structure ensures that employees can access the information required for their job functions while protecting sensitive business data from unauthorized access.

The implementation of department-specific security groups improves administrative efficiency, supports the principle of least privilege, and strengthens overall information security throughout the organisation.

Shared Folder Structure

Figure 2: Shared Folder Structure illustrates the proposed shared folder hierarchy used to organise company data and departmental resources.



Figure 2: Shared Folder Structure

Ubuntu Innovations will utilize a centralized shared folder structure to organize departmental information and support collaboration across the organisation.

The folder hierarchy includes:

- /Company/Public
- /Company/Finance
- /Company/HR
- /Company/Sales
- /Company/Software Development
- /Company/Management

Each folder stores information relevant to a specific department and is protected through access permissions assigned to security groups.

The structure improves file organisation, simplifies administration, and ensures employees can easily locate the resources required to perform their duties while maintaining appropriate security controls.

Shared Folder Access Controls

Ubuntu Innovations (Pty) Ltd will implement Role-Based Access Control (RBAC) to manage access to shared company resources. Access permissions will be assigned according to department responsibilities and business requirements.

The principle of least privilege will be applied to ensure employees only have access to the resources necessary to perform their job functions.

Access permissions are implemented using the User and Permissions Matrix

Table 10: Shared Folder Access Controls together with the shared folder hierarchy shown in **Figure 2: Shared Folder Structure**

Folder	Management	Finance	HR	Sales	Software Development
/Company/Public	Full Access	Read/Write	Read/Write	Read/Write	Read/Write
/Company/Management	Full Access	No Access	No Access	No Access	No Access
/Company/Finance	Full Access	Read/Write	No Access	No Access	No Access
/Company/HR	Full Access	No Access	Read/Write	No Access	No Access
/Company/Sales	Full Access	No Access	No Access	Read/Write	No Access
/Company/Software Development	Full Access	No Access	No Access	No Access	Read/Write

Table 10: Shared Folder Access Controls

The access control structure protects sensitive business information while ensuring employees can collaborate effectively within their departments. Access permissions are managed through security groups, simplifying administration and improving security.

Operating System Administration Guide

Ubuntu Innovations (Pty) Ltd will utilize standard Linux command-line tools to manage users, files, permissions, software installations, and system processes.

Table 11: Operating System Admin. Guide summarises the key Linux command-line utilities used to perform common system administration tasks.

Command	Purpose
pwd	Displays the current working directory.
ls	Lists files and folders within a directory.
cd	Changes the current directory.
mkdir	Creates a new directory.
touch	Creates a new file
chmod	Modifies file and folder permissions.
chown	Changes file ownership.
ps	Displays running processes.
top	Monitors system resource utilization.
apt install	Installs software packages on Ubuntu Linux.

Table 11: Operating System Admin. Guide

These commands provide system administrators with the tools required to manage operating systems, control user access, install software, and troubleshoot system issues.

Software Deployment Plan

Software Installation Procedures

Ubuntu Innovations (Pty) Ltd will follow a structured deployment process to ensure software consistency, security, and compliance across all departments.

The deployment process includes:

- Configuring Windows 11 Pro before devices are assigned to employees.
- Installing standard software applications during workstation setup.
- Deploying department-specific software according to employee roles.
- Creating company-managed user accounts and login credentials.
- Configuring cloud applications using company email accounts.
- Verifying software functionality before device allocation.
- Maintaining software inventory and licensing records.

Software Update and Patch Management

To maintain security and reliability, Ubuntu Innovations will implement the following update procedures:

- Automatic Windows updates enabled on all company devices.
- Automatic antivirus signature updates.
- Vendor-managed updates for cloud-based software.
- Immediate deployment of critical security updates.
- Removal of unsupported software.
- Scheduled compliance reviews conducted by IT support personnel.

Licensing Management

Ubuntu Innovations will maintain valid licenses for all commercial software solutions used within the organisation.

Licensed software includes:

- Windows 11 Pro
- Microsoft 365 Business
- Sage Accounting & Payroll
- BambooHR

- Canva Pro
- HubSpot CRM
- GitHub Copilot
- TeamViewer
- Jira Service Management
- Veeam Backup & Recovery

Open-source software includes:

- Visual Studio Code
- Python
- PostgreSQL
- MySQL Workbench
- Wireshark
- GitHub Desktop
- MongoDB Compass

Software Deployment Plan Conclusion

The Software Deployment Plan provides Ubuntu Innovations (Pty) Ltd with a structured approach for deploying, managing, updating, and securing software applications across all departments. Standardized software deployment improves productivity, strengthens cybersecurity, supports regulatory compliance, and ensures employees have access to the tools required to perform their roles effectively.

Business Continuity & Disaster Recovery

Business continuity planning ensures Ubuntu Innovations can continue operating during and after unexpected disruptions. The organisation relies heavily on digital systems and data to support daily operations, making backup and recovery procedures essential.

Potential risks include:

- Hardware failure
- Cyberattacks
- Power outages
- Human error
- Natural disasters
- Equipment theft

To minimise operational downtime and data loss, Ubuntu Innovations will implement a structured Backup and Disaster Recovery Plan.

Backup Strategy

Ubuntu Innovations will implement the industry-standard 3-2-1 Backup Rule.

The 3-2-1 rule requires:

- Three copies of important data.
- Two different storage locations.
- One copy stored offsite.

This approach significantly reduces the likelihood of permanent data loss while supporting business continuity objectives.

Backup Infrastructure

Network Attached Storage (NAS)

The organisation will utilise a Synology DiskStation DS1823xs+ configured with RAID 6 as the primary onsite backup repository.

Benefits include:

- Centralized storage management.
- RAID redundancy.
- Fast backup and restoration.

- Secure access control.
- Automated backup scheduling.

Cloud Backup Storage

Microsoft Azure Backup and Microsoft OneDrive for Business will be used for secure offsite backups.

Benefits include:

- Geographic redundancy.
- Disaster protection.
- Encrypted storage.
- Remote accessibility.
- Improved recovery capabilities.

Backup Schedule

Table 12: Backup Schedule summarises the organisation's backup schedule, including incremental, full, and archive backups.

Backup Type	Frequency	Storage Location	Retention Period
Incremental Backup	Daily (20:00)	NAS Device	30 Days
Full Backup	Weekly (Saturday)	NAS + Cloud Storage	12 Weeks
Archive Backup	Monthly	Cloud Storage + Encrypted External Drive	12 Months
Annual Archive	Annually	Cloud Storage	7 Years

Table 12: Backup Schedule

Daily Incremental Backups

Daily incremental backups will occur automatically at 20:00 each evening. Incremental backups copy only files that have changed since the previous backup, reducing storage requirements while ensuring recent changes are protected.

Examples of protected data include:

- Source code updates
- Customer records
- Financial transactions
- Employee documentation
- Business reports

Weekly Full Backups

A complete backup of all company systems will be performed every Saturday evening.

The backup includes:

- User profiles
- Departmental folders
- Shared storage
- Databases
- Configuration settings
- Application data

Weekly backups provide a reliable recovery point and simplify restoration procedures.

Monthly Archive Backups

Archive backups preserve historical business information for compliance and long-term retention purposes.

Examples include:

- Payroll records
- Financial statements
- Employment records
- Contracts
- Completed development projects

Offsite Cloud Storage

Offsite backup storage is a critical component of Ubuntu Innovations' disaster recovery strategy. While local NAS storage provides fast recovery capabilities, offsite cloud storage protects backup data against events that could affect the physical office environment.

Potential threats include:

- Fire
- Flooding
- Theft

- Power surges
- Hardware destruction
- Physical sabotage

Ubuntu Innovations will replicate all weekly and monthly backups to Microsoft Azure Backup and Microsoft OneDrive for Business.

Cloud Security Controls

Cloud backups will be protected through:

- AES-256 encryption
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Secure HTTPS transmission
- Backup activity monitoring
- Audit logging

The implementation of offsite cloud storage ensures that backup copies remain available even if onsite infrastructure becomes unavailable.

Recovery Objectives

Recovery objectives define the acceptable amount of downtime and data loss following a disruptive event.

Recovery Time Objective (RTO)

The Recovery Time Objective represents the maximum amount of time required to restore critical business services.

Target RTO: 4 Hours

Critical systems that must be restored within four hours include:

- File storage services
- Employee workstations
- Network connectivity
- Business applications
- Software Development environments

A four-hour recovery target minimizes operational disruption and supports business continuity.

Recovery Point Objective (RPO)

The Recovery Point Objective represents the maximum amount of data loss the organisation can tolerate.

Target RPO: 24 Hours

Because daily incremental backups are performed, Ubuntu Innovations should not lose more than one business day of information following a disaster.

This objective protects:

- Customer records
- Software Development projects
- Financial information
- Employee documentation
- Business operations

Disaster Recovery Procedures

Ubuntu Innovations will follow documented recovery procedures to restore operations quickly following a disruption.

Scenario 1: NAS Device Failure

1. Detect hardware failure through monitoring systems.
2. Replace faulty hardware components.
3. Restore latest backup from cloud storage.
4. Verify data integrity.
5. Resume normal operations.

Scenario 2: Ransomware Attack

1. Disconnect affected devices immediately.
2. Notify IT personnel.
3. Identify infected systems.
4. Remove malicious software.
5. Restore clean backups.
6. Verify functionality.
7. Return systems to production.

Scenario 3: Office Disaster

Examples include:

- Fire
- Flood
- Theft
- Extended power outage

Recovery process:

1. Obtain replacement hardware.
2. Access cloud backup systems.
3. Restore business-critical data.
4. Re-establish network connectivity.
5. Validate restored information.
6. Resume operations.

Backup Testing and Maintenance

Ubuntu Innovations will conduct quarterly backup testing to verify recovery readiness.

Testing activities include:

- File restoration testing
- Database recovery testing
- Full system recovery simulations
- Cloud backup verification
- Disaster recovery exercises

Testing results will be documented and reviewed by IT management.

Cybersecurity Framework

Ubuntu Innovations (Pty) Ltd requires a comprehensive cybersecurity framework to protect company systems, data, and users against modern cyber threats.

The framework combines preventative, detective, and corrective security controls to support confidentiality, integrity, and availability of information assets.

The cybersecurity framework consists of:

- Cybersecurity Policy
- Risk Assessment Matrix
- Incident Response Plan
- Security Awareness Training Program

Together, these controls establish a proactive security posture and support business continuity objectives.

Cybersecurity Policy

This policy establishes the minimum security standards required for all employees, devices, systems, and business applications used within Ubuntu Innovations (Pty) Ltd.

Compliance with this policy is mandatory.

Password Complexity Requirements

To reduce the risk of unauthorized access, all passwords must meet the following requirements:

- Minimum length of 14 characters.
- Include uppercase letters.
- Include lowercase letters.
- Include numbers.
- Include special characters.
- Password history of five previous passwords enforced.
- Passwords may not contain usernames or common dictionary words.

Multi-Factor Authentication (MFA)

Multi-Factor Authentication provides an additional layer of protection beyond traditional passwords.

Requirements include:

- MFA enabled for all company accounts.
- Required for email access.
- Required for VPN access.
- Required for cloud applications.
- Authenticator applications preferred over SMS authentication.

Device Encryption

All company-issued devices must use full disk encryption.

Encryption standards include:

- BitLocker (Windows devices)
- FileVault 2 (macOS devices)
- AES-256 encryption

Encryption keys will be centrally managed by IT administrators.

Antivirus and Endpoint Protection

Every endpoint device must have approved antivirus and endpoint protection software installed.

Requirements include:

- Real-time protection enabled.
- Automatic signature updates.
- Behaviour monitoring.
- Tamper protection enabled.
- Centralized monitoring by IT personnel.

Patch Management

Software vulnerabilities must be addressed through structured patch management procedures.

Deployment timelines:

Severity	Deployment Timeline
Critical	Within 48 hours
High	Within 14 days
Medium	Within 30 days
Low	Monthly Maintenance Window

Automatic updates will be enabled wherever possible.

Acceptable Use Policy

Employees must use company technology resources responsibly and ethically.

Prohibited activities include:

- Downloading illegal content.
- Installing unauthorized software.
- Sharing company credentials.
- Accessing inappropriate websites.
- Unauthorized file sharing.

Employees are required to report suspected security incidents immediately.

Risk Assessment Matrix

Risk assessment allows Ubuntu Innovations to identify potential threats and implement appropriate mitigation controls.

Risk	Likelihood	Impact	Mitigation Controls
Phishing	High	High	Security awareness training, phishing simulations, email filtering
Ransomware	Medium	High	EDR solutions, network segmentation, offsite backups
Hardware Failure	Medium	Medium	RAID storage, spare hardware, preventative maintenance
Power Outage	Medium	High	UPS systems, cloud services, backup generators
Data Leakage	Medium	High	RBAC, DLP controls, least privilege access
Unpatched Vulnerabilities	High	Medium	Vulnerability scanning, patch management procedures

Incident Response Plan

Ubuntu Innovations (Pty) Ltd requires a structured approach for responding to cybersecurity incidents. The purpose of this Incident Response Plan is to minimize damage, reduce downtime, protect company data, and restore business operations as quickly as possible following a security event.

The incident response process consists of six phases: Detection, Reporting, Containment, Eradication, Recovery, and Lessons Learned.

Detection

The first stage of incident response involves identifying potential security incidents as early as possible. Early detection helps reduce the impact of cyber threats and prevents attacks from spreading throughout the organisation.

Employees and IT personnel should monitor for indicators of compromise, including:

- Unusual network activity.
- Suspicious emails.
- Malware alerts.
- Repeated failed login attempts.
- Unauthorized system access.
- Unexpected system behaviour.

Security monitoring tools such as firewalls, antivirus software, Endpoint Detection and Response (EDR) solutions, and log management systems will assist with identifying threats.

Potential incidents must immediately be reported to the IT support team for investigation.

Reporting

Once a potential incident has been identified, it must be reported through the appropriate communication channels.

Employees must provide the following information:

- Date and time of the incident.
- Description of the issue.
- Systems affected.

- Users involved.
- Actions already taken.

The IT support team will assess the severity of the incident and determine whether escalation to management or external stakeholders is required.

Proper reporting ensures incidents are documented accurately and handled efficiently.

Containment

The objective of containment is to prevent the incident from spreading further and causing additional damage.

Containment activities may include:

- Disconnecting affected devices from the network.
- Disabling compromised user accounts.
- Blocking malicious IP addresses.
- Restricting access to affected systems.
- Isolating infected devices using network segmentation.

By limiting the scope of the incident, Ubuntu Innovations can reduce operational disruption while recovery efforts begin.

Eradication

After the incident has been contained, the root cause must be identified and removed.

Eradication activities may include:

- Removing malware or ransomware.
- Deleting malicious files.
- Closing exploited vulnerabilities.
- Applying security patches.
- Reconfiguring compromised systems.
- Resetting compromised passwords.

The objective is to ensure that the threat is completely eliminated before systems are restored to production.

Recovery

Recovery focuses on restoring affected systems and services to normal operation.

Recovery activities include:

- Restoring data from verified backups.
- Rebuilding affected systems.
- Reconnecting devices to the network.
- Testing system functionality.
- Monitoring systems for signs of recurring threats.

Critical business services should be restored according to the organisation's Recovery Time Objective (RTO) and Recovery Point Objective (RPO).

The recovery process ensures business operations can resume safely and efficiently.

Lessons Learned

Following every incident, Ubuntu Innovations will conduct a formal post-incident review.

The purpose of the review is to:

- Identify the root cause of the incident.
- Evaluate the effectiveness of the response.
- Identify weaknesses in existing controls.
- Recommend improvements.
- Update policies and procedures where necessary.

Lessons learned from previous incidents help strengthen the organisation's overall cybersecurity posture and improve future response efforts.

Incident Response Plan Summary

The Incident Response Plan provides Ubuntu Innovations (Pty) Ltd with a structured framework for detecting, responding to, containing, and recovering from cybersecurity incidents. By following a consistent response process, the organisation can reduce operational disruption, protect business-critical information, and improve overall resilience against evolving cyber threats.

Security Awareness Training Program

Ubuntu Innovations (Pty) Ltd recognises that employees represent the first line of defence against cybersecurity threats. While technical controls such as firewalls, antivirus software, and access controls are essential, employee awareness remains critical in preventing security incidents.

The purpose of the Security Awareness Training Program is to educate employees on common cyber threats, promote secure working practices, and ensure that all staff understand their responsibilities in protecting company information assets.

Training will be conducted during employee onboarding and reinforced through scheduled awareness sessions throughout the year.

Training Module 1: Phishing Awareness

Phishing remains one of the most common attack methods used by cybercriminals. This training module focuses on helping employees identify and report suspicious communications.

Topics covered include:

- Identifying phishing emails.
- Recognising suspicious hyperlinks.
- Understanding social engineering tactics.
- Reporting phishing attempts.
- Email security best practices.

Training Frequency: Quarterly

Training Module 2: Password Security and Identity Protection

Strong authentication practices reduce the risk of unauthorized access to company systems.

Topics covered include:

- Creating secure passwords and passphrases.
- Password management best practices.
- Multi-Factor Authentication (MFA).
- Credential theft prevention.
- Account security responsibilities.

Training Frequency: Quarterly

Training Module 3: Safe Internet Usage

Employees must understand how to use internet resources safely and responsibly.

Topics covered include:

- Safe web browsing.
- Identifying malicious websites.
- Secure use of public Wi-Fi.
- VPN usage.
- Downloading software safely.

Training Frequency: Bi-Annually

Training Module 4: Device and Physical Security

Company devices contain sensitive business information and must be protected from theft, damage, and unauthorized access.

Topics covered include:

- Device locking procedures.
- Physical security best practices.
- Secure storage of company equipment.
- Removable media risks.
- Mobile device security.

Training Frequency: Bi-Annually

Training Module 5: Incident Reporting

Employees must understand how to respond when they suspect a security incident has occurred.

Topics covered include:

- Recognising security incidents.
- Internal reporting procedures.
- Escalation processes.
- Immediate response actions.
- Documentation requirements.

Training Frequency: Annually

Training Program Summary

The Security Awareness Training Program helps create a culture of cybersecurity awareness throughout Ubuntu Innovations. By educating employees on emerging threats and secure working practices, the organisation can reduce risk, strengthen compliance, and improve overall security resilience.

Module	Frequency
Phishing	Quarterly
Password Security	Quarterly
Safe Internet Usage	Bi-Annually
Device Security	Bi-Annually
Incident Reporting	Annually

Technical Support Operations

Ubuntu Innovations (Pty) Ltd requires documented troubleshooting procedures to ensure common IT issues can be resolved efficiently and consistently. Standardised troubleshooting guides help minimise downtime, improve service quality, and support operational continuity.

The following troubleshooting scenarios represent common incidents likely to occur within the organisation.

Workstation Power Failure

Issue Description

A workstation fails to power on or unexpectedly shuts down during normal operation.

Possible Causes

- Power cable disconnected.
- Faulty power supply unit.
- Damaged power outlet.
- Battery failure.
- Hardware malfunction.

Troubleshooting Procedure

1. Verify that the power cable is securely connected.
2. Confirm that the wall outlet is functioning correctly.
3. Test the workstation using an alternative power outlet.
4. Inspect the power adapter or power supply unit for damage.
5. Remove external peripherals and attempt startup.
6. Check battery status if using a laptop.
7. Escalate to hardware diagnostics if the issue persists.

Resolution

Replace faulty power components or hardware as required and verify successful system startup.

Network Connectivity Failure

Issue Description

An employee is unable to access the internet, shared resources, or network services.

Possible Causes

- Faulty network cable.
- Wireless connectivity issues.
- DHCP configuration problems.
- Switch port failure.
- Router or firewall malfunction.

Troubleshooting Procedure

1. Confirm whether other users are experiencing the same issue.
2. Check physical network connections.
3. Verify Wi-Fi connectivity status.
4. Restart the affected workstation.
5. Use ipconfig or ifconfig to verify IP configuration.
6. Ping the default gateway.
7. Test internet connectivity.
8. Inspect switch and router status indicators.
9. Escalate to network administration if necessary.

Resolution

Restore connectivity through cable replacement, configuration correction, or network equipment repair.

Shared Printer Failure

Issue Description

Employees are unable to print documents using the shared network printer.

Possible Causes

- Printer offline.

- Paper jam.
- Low toner levels.
- Print spooler errors.
- Network communication failure.

Troubleshooting Procedure

1. Verify printer power status.
2. Check for error messages.
3. Confirm paper and toner availability.
4. Verify network connectivity.
5. Restart the printer.
6. Restart the print spooler service.
7. Reinstall the printer driver if required.
8. Submit a test print job.

Resolution

Resolve hardware or configuration issues and confirm successful printing functionality.

Project Outcomes & Recommendations

Project Outcomes

The Ubuntu Innovations IT Infrastructure Project successfully delivered a comprehensive set of technical solutions designed to support the organisation's operational, security, and growth requirements.

Key project outcomes include:

Improved Infrastructure Planning

A complete hardware and software solution was developed to support the organisation's 25 employees while allowing for future expansion. The proposed infrastructure provides improved reliability, scalability, and performance.

Enhanced Network Design

A structured network architecture was created to support secure communication between employees, servers, storage systems, printers, and internet services. The design incorporates centralized management and improved network security controls.

Stronger Access Management

Role-based access control (RBAC) was implemented through department-specific security groups and shared folder permissions. This approach ensures employees can access only the resources required to perform their duties.

Improved Business Continuity

The Backup and Disaster Recovery Plan provides mechanisms for protecting business-critical information and recovering from disruptions. The implementation of daily, weekly, and monthly backups significantly reduces the risk of permanent data loss.

Increased Cybersecurity Readiness

The Cybersecurity Framework establishes policies and procedures that strengthen organisational security. Password policies, multi-factor authentication, device encryption, patch management, and employee awareness training collectively reduce cybersecurity risks.

Standardised Technical Support Procedures

Documented troubleshooting guides provide IT personnel with repeatable processes for resolving common technical issues quickly and efficiently.

Career Readiness Software Development

The project enabled team members to develop technical documentation, project management, communication, presentation, and professional development skills that align with industry expectations for entry-level IT Support professionals.

Recommendations

To maximise the long-term effectiveness of the proposed solution, the following recommendations are provided:

Implement Network Segmentation

As the organisation grows, Virtual Local Area Networks (VLANs) should be introduced to separate departmental traffic and improve network security.

Adopt Centralised Identity Management

Implementing Microsoft Entra ID (Azure Active Directory) would simplify user administration, strengthen access control, and improve authentication security.

Conduct Regular Security Audits

Quarterly security assessments should be performed to identify vulnerabilities and ensure compliance with company security policies.

Expand Backup Testing

Regular disaster recovery exercises should be conducted to verify that backup systems remain effective, and recovery objectives can be achieved.

Increase Security Awareness Training

Cybersecurity awareness sessions should be updated regularly to address emerging threats such as ransomware, phishing, and social engineering attacks.

Monitor Infrastructure Growth

The organisation should regularly review storage capacity, network utilisation, and hardware performance to ensure the infrastructure continues to meet business demands.

Develop Future Automation Solutions

Automation technologies such as PowerShell scripting, ticketing workflows, and monitoring tools should be explored to improve operational efficiency and reduce administrative overhead.

Conclusion

This project provided a comprehensive assessment of the infrastructure requirements for Ubuntu Innovations (Pty) Ltd and resulted in the development of a complete IT support solution aligned with industry best practices.

The proposed solution addresses key business requirements by providing reliable hardware, secure networking, structured access management, business continuity planning, cybersecurity controls, and technical support procedures. Each recommendation was selected to improve organisational efficiency, strengthen security, and support future growth.

The project also demonstrated the practical application of knowledge gained through the Google IT Support Professional Certificate and Tech Career Accelerator programme. Areas including networking, operating systems, system administration, cybersecurity, troubleshooting, and professional communication were integrated into a realistic business scenario.

By implementing the recommendations contained within this report, Ubuntu Innovations will be better positioned to maintain secure operations, minimise downtime, protect business information, and support long-term organisational growth.

The project demonstrates the importance of strategic IT planning and highlights the critical role IT support professionals play in maintaining modern business environments.

References

Course References

Google. (2024). *Google IT Support Professional Certificate*. Coursera.

Tech Career Accelerator. (2025). *IT Support Capstone Project Guide*.

Technical References

Microsoft Learn. (2025). *Microsoft Entra ID Documentation*.

Microsoft Learn. (2025). *Windows Security and BitLocker Documentation*.

CompTIA. (2024). *CompTIA Security+ Study Guide*.

Cisco Networking Academy. (2024). *Introduction to Networks*.

Synology. (2025). *DiskStation DS1823xs+ Product Documentation*.

Ubiquiti. (2025). *UniFi Network Platform Documentation*.

Ubuntu Documentation Team. (2025). *Ubuntu Server Administration Guide*.

National Institute of Standards and Technology (NIST). (2024). *Cybersecurity Framework (CSF) 2.0*.

Microsoft. (2025). *Microsoft 365 Business Documentation*.
